

Measuring Data Exposure in LLM-Integrated Productivity Tools

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Project Focus

This project compares how user data exposure differs across LLM-integrated productivity tools during controlled use.

Problem Statement

LLM-integrated productivity tools interact with user data through browser, editor, and system-level interfaces during normal use, yet there is no systematic comparison of how user data is exposed to external servers across different tools. This makes it difficult to assess and compare their privacy impact in a consistent way.

Research Question

How does user data exposure differ across LLM-integrated productivity tools during controlled use?

Objectives

- 1 Define a controlled setup for evaluating LLM-integrated productivity tools
- 2 Measure user data exposure during tool usage
- 3 Compare exposure across tools under the same conditions
- 4 Summarize results per tool with a clear exposure assessment and confidence level

Scope

The study focuses on a small set of LLM-integrated productivity tools tested under identical controlled tasks. The analysis is limited to observable outbound data exposure. Tools will be selected to represent different types of integration, such as browser-based tools, editor-based tools, and web interfaces.

Methodology

The study will use a controlled experimental setup in which each selected tool is evaluated using the same tasks and identical input data. Outbound network activity will be observed during execution, and baseline runs without the tool enabled will be used for comparison. Data exposure will be assessed using consistent indicators, including whether known input data appears in transmitted content, the volume of outbound data, the number of outbound requests, and the number of external domains contacted. The results will then be compared across tools under the same conditions.

Implementation and Output

The implementation will use Python together with standard libraries for data processing and analysis. Outbound traffic will be observed using mitmproxy, and experiments will be conducted in a browser environment. Captured data will be stored in structured formats such as CSV or JSON. The project will produce a comparative analysis of data exposure across selected tools. For each tool, the results will be summarized in a clear statement describing the level of data exposure under the tested conditions, together with a confidence assessment based on the consistency and completeness of the observations.

Limitations

The study focuses only on observable outbound traffic under controlled conditions. Some traffic may not be accessible due to encryption or other restrictions, and this will be noted as a limitation.